

Erdős Problem #709

EP-709

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $f(n)$ be minimal such that, for any $A = \{a_1, \dots, a_n\} \subseteq [2, \infty) \cap \mathbb{N}$ of size n , in any interval I of $f(n) \max(A)$ consecutive integers there exist distinct $x_1, \dots, x_n \in I$ such that $a_i \mid x_i$.

Obtain good bounds for $f(n)$, or even an asymptotic formula.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>